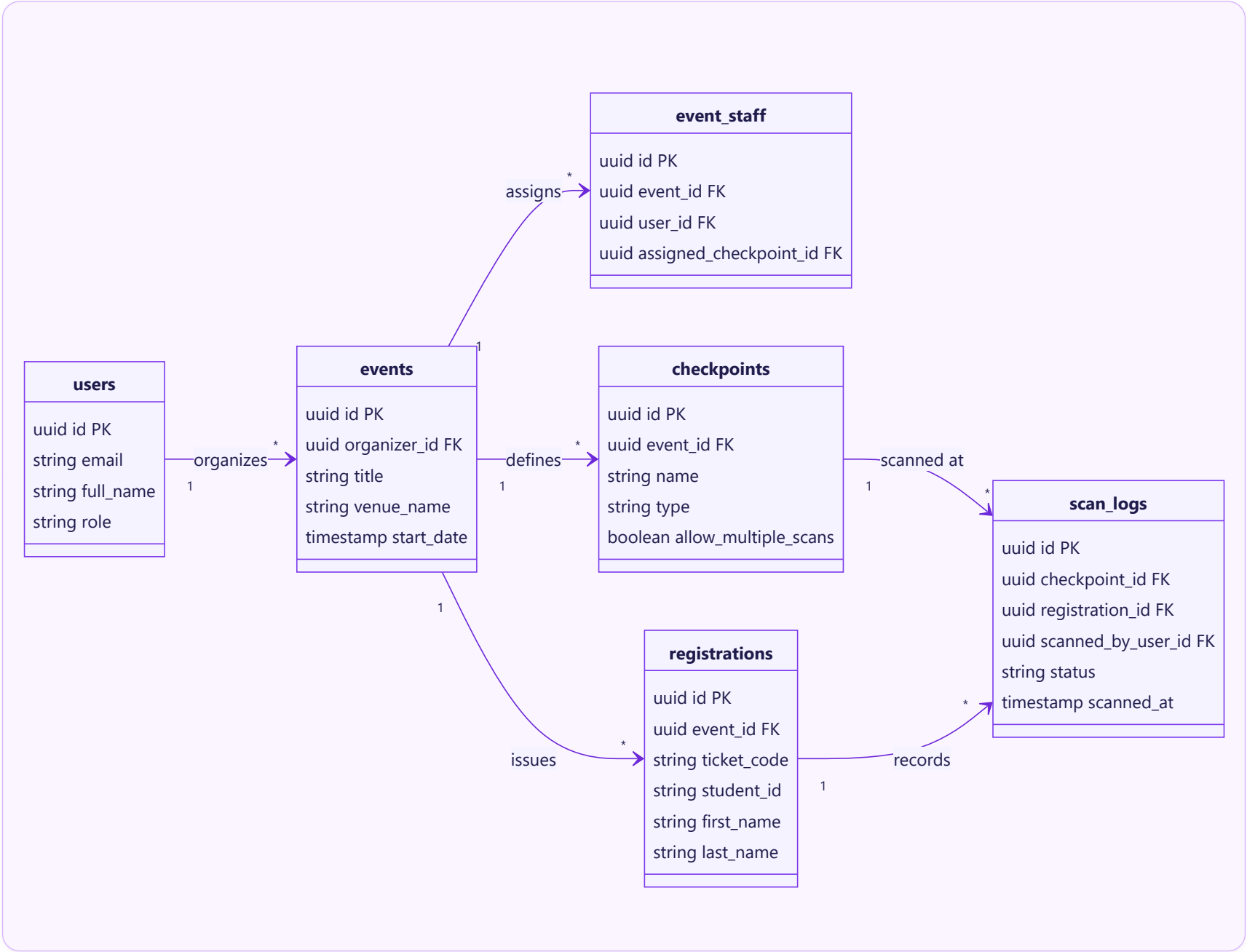


Checkpoint Database Design

Entity Relationship Diagram



1. Table: users

Stores accounts for organizers, event managers, and check-in staff volunteers. Attendees do not have accounts in this table.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key identifying the user account.
email	VARCHAR(255)	UNIQUE, NOT NULL	Email address used for authentication.
password_hash	VARCHAR(255)	NOT NULL	Cryptographically hashed password (e.g., bcrypt/argon2).
full_name	VARCHAR(150)	NOT NULL	Full legal or display name of the staff or organizer.
role	user_role (ENUM)	NOT NULL	Access control role: ADMIN , ORGANIZER , or STAFF .
is_active	BOOLEAN	NOT NULL	Account status flag. If false, user cannot authenticate.
created_at	TIMESTAMPTZ	NOT NULL	Timestamp when the account was registered.
updated_at	TIMESTAMPTZ	NOT NULL	Timestamp when the user details were last modified.

2. Table: events

Stores academic seminars, conferences, and expos managed by organizers.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key identifying the event.
organizer_id	UUID	FK, NOT NULL	References users(id) . Identifies the event owner.
title	VARCHAR(255)	NOT NULL	Official title of the event.
slug	VARCHAR(120)	UNIQUE, NOT NULL	URL-safe string for public ticket lookup and walk-in forms.
description	TEXT	NULLABLE	Overview, agenda, or instructions for the event.
venue_name	VARCHAR(255)	NOT NULL	Physical location, auditorium, or campus building.
start_date	TIMESTAMPTZ	NOT NULL	Official starting date and time of the event.
end_date	TIMESTAMPTZ	NOT NULL	Official ending date and time of the event.
is_active	BOOLEAN	NOT NULL	Controls whether the event is active or archived.
metadata	JSONB	NOT NULL	Extensible settings (e.g., max capacity, tags, banners).
created_at	TIMESTAMPTZ	NOT NULL	Timestamp when the event was created.
updated_at	TIMESTAMPTZ	NOT NULL	Timestamp when event details were last updated.

3. Table: checkpoints

Defines the distinct physical scan points (gates, breakout rooms, sponsor booths) within an event.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key for the checkpoint.
event_id	UUID	FK, NOT NULL	References events(id) . The event this checkpoint belongs to.
name	VARCHAR(150)	NOT NULL	Descriptive station name (e.g., "Main Gate A", "Booth #04").
type	checkpoint_type (ENUM)	NOT NULL	Type: MAIN_ENTRY , SESSION , BOOTH , or EXIT .
description	TEXT	NULLABLE	Location notes, exhibitor info, or usher directions.
allow_multiple_scans	BOOLEAN	NOT NULL	If true, permits repeat scans (for booths). If false, rejects duplicates (for entry).
is_active	BOOLEAN	NOT NULL	Toggle to enable or disable scanning at this specific station.
created_at	TIMESTAMPTZ	NOT NULL	Timestamp when checkpoint was added.
updated_at	TIMESTAMPTZ	NOT NULL	Timestamp when checkpoint was last updated.

4. Table: event_staff

Associates staff volunteers with specific events and optionally restricts them to a particular checkpoint.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key for the assignment record.
event_id	UUID	FK, NOT NULL	References events(id) . Event the staff member is assigned to.
user_id	UUID	FK, NOT NULL	References users(id) . The staff user assigned.
assigned_checkpoint_id	UUID	FK, NULLABLE	References checkpoints(id) . Designated station for the staff member.
created_at	TIMESTAMPTZ	NOT NULL	Timestamp when staff assignment was created.

5. Table: registrations

Represents an attendee's digital pass or badge. Generated via CSV import or on-site walk-in form. Attendees require no passwords.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key for the attendee registration.
event_id	UUID	FK, NOT NULL	References events(id) . The event the attendee is registered for.
ticket_code	VARCHAR(64)	UNIQUE, NOT NULL	Unique cryptographic token encoded inside the QR pass.

Field Name	Data Type	Constraints	Description
first_name	VARCHAR(100)	NOT NULL	Attendee's given name.
last_name	VARCHAR(100)	NOT NULL	Attendee's family name.
email	VARCHAR(255)	NOT NULL	Email address where the pass and updates are delivered.
student_or_id_number	VARCHAR(100)	NULLABLE	Student ID or Employee ID for academic attendance validation.
affiliation	VARCHAR(255)	NULLABLE	College department, organization, or company.
ticket_type	VARCHAR(50)	NOT NULL	Attendee tier (e.g., STUDENT , VIP , SPEAKER , REGULAR).
status	registration_status	NOT NULL	Status: CONFIRMED , CANCELLED , or PENDING .
first_checked_in_at	TIMESTAMPTZ	NULLABLE	Timestamp of the attendee's very first successful event check-in.
custom_data	JSONB	NOT NULL	Additional fields (e.g., dietary choices, workshop selection).
created_at	TIMESTAMPTZ	NOT NULL	Timestamp of initial registration or CSV import.
updated_at	TIMESTAMPTZ	NOT NULL	Timestamp when registration details were last modified.

6. Table: scan_logs

An append-only audit trail logging every scan event across all checkpoints.

Field Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique primary key identifying the scan record.
checkpoint_id	UUID	FK, NOT NULL	References checkpoints(id) . Station where scan occurred.
registration_id	UUID	FK, NOT NULL	References registrations(id) . Attendee pass scanned.
scanned_by_user_id	UUID	FK, NULLABLE	References users(id) . Staff member who conducted the scan.
status	scan_status (ENUM)	NOT NULL	Outcome of scan: SUCCESS , DUPLICATE , or REJECTED .
scanned_at	TIMESTAMPTZ	NOT NULL	Exact timestamp when the scan was executed.
device_info	JSONB	NOT NULL	Client telemetry (browser user-agent, scanner device metadata).
notes	TEXT	NULLABLE	Optional usher remarks or exception notes.